

Block Coding with Bees

Substitute Snapshot (Print this page)

Goal

Get the  bee from start to  hive while visiting all  flowers and avoiding  pesticide, using at least one loop.

Time

- 10 min: Ignite Thinking (pictures + notice/wonder)
- 30–35 min: Work Time (review arrows → block coding + loops)
- 10 min: Reflection (group process + debugging)

Board/Slide Directions

- “Use arrows first. Then convert to blocks and add a loop.”
- “Run your program. If it fails, fix the bug and try again.”

Links (QR + click)

[Lesson slides](#)

[Grid Image Cards](#)

[Coding Blocks](#)



Visual Legend (use while teaching)



Bee (student)



Hive (goal)



Flower (must visit)



Pesticide (avoid)

Lesson Overview

Pillar	Computational Thinking
Grade	2–3
Group Size	2–3
Essential Questions	• How can loops and block coding be used to simplify an algorithm? • What is block coding? • How does block coding work? • What is a loop?
Criteria for Success	Code the bee to get it from its starting point to the beehive. Use block coding to communicate the bee's route to others.
Constraints	• The bee must pollinate all of the flowers. • The bee must avoid the pesticide. • Use at least 1 loop in your code.

Learning Targets (Standards)

- I can decompose a problem into smaller steps, in multiple ways (2–3.CT.4).
- I can identify the essential information needed to get the Bee to the hive (2–3.CT.5).
- I can code using fixed and changing pieces of information (2–3.CT.7).
- I can document my plan to get the bee to the hive using block-coding (2–3.CT.10).

CASEL Competencies / SEL Standards

Habit 2: Begin with the End in Mind

- Set a short-term goal and begin working towards it (1c.2a).
- Identify steps in reaching the goal (1c.2b).

Habit 5: Seek First to Understand, then Be Understood

- Identify verbal, physical, and situational cues that indicate how others may feel (2A.2a).
- Communicate perceived understanding of the expressed feelings and perspectives of others (2A.2b).

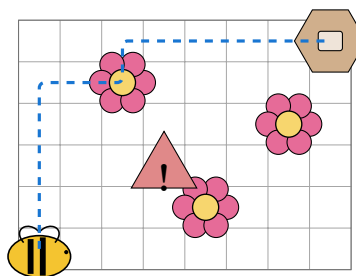
Materials

- Lesson slides: [Google Slides](#)
- Scissors
- Pencils
- 1 set per group of each of the following:
 - Grid (large poster floor grid or painter’s tape grid)
 - Arrow cards from previous floor coding lessons
 - Grid Image Cards: [Google Drive file](#)
 - Coding Blocks: [Google Drive file](#)

Sub-Friendly Prep Checklist

- ☐ Put up the floor grid (or tape grid) before students arrive.
- ☐ Place Bee/Hive/Flower/Pesticide picture cards (or show the grid image on the board).
- ☐ Give each group arrow cards and block-coding cards.
- ☐ Decide a “driver” and “checker” role (swap halfway).

Mini Visual: Example Grid (replace with your slide image)



Tip: Use this diagram only as a teaching aid. Students should match the official grid image from the slides/smartboard.

Lesson Structure

Phase (Time)	Plan
Ignite Thinking (10 mins)	• Show students images of bees: – Image 1: Bee pollinating a flower – Image 2: Danger of pesticides • Ask students what they notice and know about these pictures. • Tell students they will use coding knowledge to help bees navigate through an ecosystem full of flowers and pesticides. • Review learning targets.
Work Time (30–35 mins)	1. Task 1: Arrow Coding Review (~5–7 minutes) Have students set up their floor grids to match the image on the smartboard. Students work in their groups using arrow cards to get the bee to the hive (keeping constraints in mind). 2. Introduce Block Coding (~5 minutes) Introduce block coding and describe what it is. Ask how often they had a step/arrow repeat; introduce the vocabulary term <i>loop</i>. Review essential questions and vocabulary terms. 3. Task 2: Turn Arrow Coding into Block Coding (~5 minutes) Whole-group: Model changing arrow codes into blocks with loops. Introduce Run and End commands. 4. Task 3: Block Coding (~15–20 minutes) Students set up their floor grids to match the image on the smartboard. Students work in their groups using block coding to get the bee to the hive. If time, have groups switch places and run each other's program to see if it works or if there are bugs.
Reflection (10 mins)	Use look-fors/documentation during discussion: • How did you feel about block coding? • What went well in your group? • How did you make sure that everyone's perspectives were heard in your group? • Anything from the look-fors?

Look Fors / Document

- Students using talking points, actively listening to each other, and taking turns/hearing other perspectives.
- Groups that handle conflict well.
- Examples of programs using block coding accurately or with bugs that need to be fixed.

Common Sticking Points (quick fixes)

- **Missed a flower:** Have students highlight/check off each  flower as they “visit” it.
- **Hit pesticide:** Ask, “Which step took you onto ?” Remove or change that step.
- **Loop confusion:** Identify the repeating pattern first, then wrap it in a loop (e.g., repeat 3 times).
- **Group conflict:** Use roles (driver/checker) and a 30-second “listen first” turn.